

A necessary inverse-dimension floor in endpoint balance

Research note; independently reviewed

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The quantitative endpoint statements concern monotone columns satisfying

$$\mathbb{E}_w \prod_{i \in S} p_i = \frac{1}{|S| + 1} \quad (S \subseteq [m]), \quad u_i = 1 - p_i, \quad s = \sum_i u_i.$$

They must allow a nonzero additive error even arbitrarily near the endpoint. We first give an exact tensor example and then transfer its asymptotic inequalities to a fully permutation-fair family of finite uniform dice.

Proposition 1. *For every $m \geq 2$ there is a finite positive-weight monotone tensor satisfying the displayed moments and having a last row with*

$$s = \max_i u_i \asymp m^{-2}, \quad \max_i |mu_i - s| \asymp m^{-1}.$$

Consequently a uniform endpoint-profile error $o(m^{-1})$ is impossible in this class. Neither $\max_i mu_i \leq Cs$ nor $\max_i mu_i \leq C(s + o(m^{-1}))$ can hold with an absolute constant.

Proof. Let $r \in \{m, m+1\}$ be odd, let P_r be the Legendre polynomial normalized by $P_r(1) = 1$, and set

$$\varepsilon = \frac{1}{2r(r+1)}, \quad p_1(t) = t - \varepsilon P_r(2t-1), \quad p_i(t) = t \quad (2 \leq i \leq m).$$

The classical bounds $|P_j(x)| \leq 1$ and $|P'_r(x)| \leq r(r+1)/2$ on $[-1, 1]$ give $p'_1(t) \geq 1/2$. For completeness, the first bound follows from the integral identity

$$P_j(x) = \frac{1}{\pi} \int_0^\pi (x + i\sqrt{1-x^2} \cos \phi)^j d\phi;$$

the integrand has modulus at most one. The derivative expansion $P'_r = \sum_{0 \leq j < r, r-j \text{ odd}} (2j+1)P_j$, obtained by telescoping $P'_{j+1} - P'_{j-1} = (2j+1)P_j$, proves the second bound. These identities also follow directly from the Legendre generating function. Since r is odd, $p_1(0) = \varepsilon$ and $p_1(1) = 1 - \varepsilon$. Thus every p_i is increasing and takes values in $[0, 1]$.

Under uniform measure on $[0, 1]$, a product over k columns is either t^k or $t^k - \varepsilon t^{k-1} P_r(2t-1)$. Orthogonality of the shifted Legendre polynomial to all degrees below r makes its integral $1/(k+1)$ in both cases, since $k-1 \leq m-1 < r$.

We record an elementary positive quadrature with both endpoints to make the example finite. For an integer $K \geq 3$, let J be the degree- $(K-2)$ orthogonal polynomial for weight $t(1-t)$ on $[0, 1]$. Its roots are simple and interior. Take these roots and the two endpoints as nodes, with interpolatory weights. Polynomial division by $t(1-t)J(t)$ shows that this rule is exact through degree $2K-3$: the quotient has degree at most $K-3$, and its integral against $t(1-t)J$ is zero by orthogonality. All weights are positive. For an interior root a , apply exactness to $t(1-t)(J(t)/(t-a))^2$; only the node a contributes, and the integral is strictly positive. For the endpoints use $(1-t)J(t)^2$ and $tJ(t)^2$, respectively.

Choose $K = r + 2$. Every product of the columns has degree at most $r + m - 1 \leq 2r - 1$, so this positive rule preserves all the required moments. Evaluate the columns at its ordered nodes. At its last node $t = 1$ we have $u_1 = \varepsilon$ and $u_i = 0$ for $i > 1$. Hence $s = \varepsilon$ and $mu_1 - s = (m - 1)\varepsilon$, which are the claimed orders. The asserted failures of uniform bounds follow immediately. $\square$

Corollary 1 (Actual finite uniform dice). *For all sufficiently large m there is a permutation-fair family of $m + 1$ finite uniform dice and a face of one die whose comparison row satisfies*

$$s \asymp m^{-2}, \quad \max_i |m(1 - p_i) - s| \asymp m^{-1}.$$

Thus the inverse-dimension additive floor is necessary even for actual finite uniform dice. No face-count bound is asserted.

Proof. Keep r, ε from the preceding proof. Define a probability measure σ on $[0, 1]$ with mass ε at each endpoint and interior density

$$f(t) = 1 - 2\varepsilon P'_r(2t - 1).$$

The preceding derivative bound gives $1/2 \leq f \leq 3/2$, and its interior CDF is $F_\sigma(t) = t - \varepsilon P_r(2t - 1)$. For $1 \leq j \leq m$, integration by parts and Legendre orthogonality give

$$\int x^j d\sigma(x) = 1 - j \int_0^1 t^{j-1} F_\sigma(t) dt = \frac{1}{j+1}.$$

Take $t = 1 - \varepsilon/(100m)$. Since $f \leq 3/2$, $1 - F_\sigma(t) = \varepsilon + O(\varepsilon/m)$. Use `finite_profile_realization.tex` with $\rho = \varepsilon/(100m)$. At the resulting old-die face,

$$1 - p_1 = \varepsilon + O(\varepsilon/m), \quad 0 \leq 1 - p_i \leq \frac{\varepsilon}{50m} \quad (i \geq 2).$$

Consequently $s \asymp \varepsilon \asymp m^{-2}$ and $m(1 - p_1) - s \asymp m\varepsilon \asymp m^{-1}$. The upper bound for every column follows from the same inequalities. $\square$

The Legendre derivative identities are also recorded in the NIST Digital Library of Mathematical Functions, Section 18.9. The proof above supplies the estimates used here. This example does not contradict a quadratic lower bound on equal row counts or on the number of faces of an individual die.